

Charcot-Marie-Tooth Disease Dominant Intermediate E (CMTDIE): A Comprehensive Disease Characteristics Report

Disease: Charcot-Marie-Tooth Disease Dominant Intermediate E **MONDO ID:** MONDO:0013758 | **OMIM:** #614455 | **Category:** Mendelian (autosomal dominant) **Causal gene:** *INF2* (Inverted Formin 2), 14q32.33, HGNC:23791

Summary

Charcot-Marie-Tooth disease Dominant Intermediate E (CMTDIE) is a rare autosomal-dominant Mendelian disorder that uniquely couples an **intermediate-type peripheral neuropathy** with **focal segmental glomerulosclerosis (FSGS)** of the kidney. It is caused by heterozygous mutations in *INF2*, a gene encoding an endoplasmic-reticulum-anchored, actin-nucleating formin. Pathogenic variants cluster tightly in the **diaphanous-inhibitory domain (DID)** encoded by exons 2–4. In a landmark cohort, 12 of 16 (75%) patients with CMT plus glomerulopathy carried DID mutations, establishing *INF2* as the dominant cause of the dual phenotype ([PMID: 22187985](#)).

The mechanism is a **gain-of-function actinopathy**, not haploinsufficiency: point-mutant *Inf2* knock-in mice develop glomerular disease while *Inf2* knockouts do not ([PMID: 39536114](#)). Loss of the DID-mediated autoinhibition produces **excessive/dysregulated actin polymerization** that injures two highly polarized cell types simultaneously: **Schwann cells**, producing a "Schwann-cell actinopathy" with demyelinating-plus-axonal features and intermediate nerve conduction velocities ([PMID: 24487800](#)), and **podocytes**, producing FSGS via dysregulated dynein-mediated trafficking of nephrin to the proteasome, abnormal mitochondrial dynamics, and terminal MRTF/SRF- and p53-driven cell death ([PMID: 39621430](#), [PMID: 39586895](#)).

There is a **positional genotype-phenotype gradient** along the DID: N-terminal residues (57–184) produce the dual CMT/FSGS phenotype, whereas more C-terminal residues (184–245) tend to produce isolated FSGS (PMID: 37491439). Renal disease is progressive, frequently reaching end-stage renal disease (ESRD) in the third-to-fourth decade, but—importantly for counseling—**genetic (INF2) FSGS does not recur after kidney transplantation**, unlike idiopathic FSGS (PMID: 27733133). No disease-specific therapy exists; management is supportive, though proteasome inhibition and allele-selective silencing are promising experimental strategies.

Section 1: Disease Information

Overview. CMTDIE is a hereditary neurologic-renal syndrome in which an intermediate form of Charcot-Marie-Tooth peripheral neuropathy co-occurs with steroid-resistant FSGS. "Intermediate" refers to nerve conduction velocities that fall between the demyelinating (CMT1, <38 m/s) and axonal (CMT2, >45 m/s) ranges, reflecting mixed demyelinating and axonal pathology. The disease is distinctive among the CMTs because the same mutation damages both the peripheral nervous system and the kidney glomerulus.

Key identifiers.

Resource	Identifier
MONDO	MONDO:0013758
OMIM	#614455 (Charcot-Marie-Tooth disease, dominant intermediate E)
Gene	INF2, OMIM *610982, HGNC:23791, 14q32.33
ICD-10	G60.0 (Hereditary motor and sensory neuropathy)
MeSH	Charcot-Marie-Tooth Disease (D002607)
Orphanet	Related entry: Charcot-Marie-Tooth disease with glomerulopathy

Synonyms / alternative names. CMTDIE; Dominant intermediate Charcot-Marie-Tooth disease type E (DI-CMTE); CMT–FSGS; INF2-related CMT with glomerulopathy; hereditary neuropathy with glomerulopathy.

Information source. The knowledge base is derived from **aggregated disease-level resources** (OMIM, ClinVar) plus **individual-patient case series and family pedigrees** reported in the primary literature. There is no large EHR-derived cohort; the disease is rare, and knowledge rests on multi-generation families and small case series.

Section 2: Etiology

Primary cause — genetic. CMTDIE is a monogenic disorder caused by **heterozygous mutations in *INF2***. In the defining cohort, Boyer et al. genotyped 16 index patients with CMT plus FSGS who lacked *PMP22/MPZ* mutations and identified nine novel heterozygous *INF2* mutations in 12/16 (75%), all in exons 2–3 encoding the DID ([PMID: 22187985](#)). Inheritance is autosomal dominant; *de novo* mutations also occur and may be relatively common in the dual phenotype ([PMID: 24174593](#)).

Genetic risk factors. The causal variants are the DID missense/in-frame variants themselves. The *position* of the variant is the principal modifier of the phenotype (see Section 4). No independent susceptibility loci or GWAS signals are described—this is a Mendelian, not complex, disease.

Environmental risk factors. None established as causative. CMTDIE is fully determined by the germline *INF2* variant. However, experimental models show that a "second hit" of glomerular stress unmasks the renal phenotype: R218Q knock-in mice are normal at baseline but develop proteinuria/FSGS after puromycin aminonucleoside (PAN) or protamine sulfate injury ([PMID: 38915495](#), [PMID: 27350175](#)). This implies that mutant *INF2* confers a susceptibility to injury rather than causing spontaneous glomerular destruction, and that podocyte stressors could plausibly modulate human disease onset—though this is inferred from models, not demonstrated clinically.

Protective factors. No genetic or environmental protective factors are established. The observation that *INF2* knockout does not cause disease implies that reducing mutant allele expression (allele-selective silencing) would be protective—a therapeutic hypothesis, not a natural protective factor.

Gene-environment interactions. Not characterized in humans. The model data (mutation + injury synergy) are the closest analog.

Section 3: Phenotypes

CMTDIE has two organ-system phenotype clusters: **neurologic** and **renal**, plus occasional additional features.

Neurologic phenotypes

Phenotype	Type	HPO term	Onset	Severity/ Progression
Distal muscle weakness (legs > arms)	Clinical sign	HP:0009053 (distal lower limb amyotrophy)	Childhood–adolescence (mean onset ~11.5 y)	Slowly progressive
Peripheral sensory loss	Symptom	HP:0106487	Childhood–adult	Progressive
Pes cavus	Physical manifestation	HP:0001761	Childhood	Stable/ progressive
Distal muscle atrophy	Clinical sign	HP:0003693	Adolescence	Progressive
Kyphoscoliosis	Physical manifestation	HP:0002751	Variable	Variable
Intermediate nerve conduction velocity	Laboratory/ electrophysiology	HP:0030181 (variable NCV)	Detectable at diagnosis	Stable trait

Case reports document mean CMT onset ~11.5 years (range 3–17) with slowly progressive sensorimotor polyneuropathy, pes cavus, and kyphoscoliosis ([PMID: 30680856](#)). Rare families show additional CNS features (intellectual disability, more severe sensorineural hearing loss) ([PMID: 24174593](#)) or transient speech difficulty ([PMID: 25943269](#)).

Renal phenotypes

Phenotype	Type	HPO term	Onset	Severity/ Progression
Focal segmental glomerulosclerosis	Pathology/lab	HP:0000097	Childhood–adulthood	Progressive to ESRD
Proteinuria	Laboratory abnormality	HP:0000093	Childhood–adult	Progressive
Nephrotic syndrome	Clinical	HP:0000100	Variable	Steroid-resistant
End-stage renal disease	Clinical	HP:0003774	3rd–4th decade typically	Terminal renal outcome

Renal involvement ranges from minimal proteinuria to steroid-resistant nephrotic syndrome progressing to ESRD. Some *INF2* mutations produce **isolated CMT with minimal/absent kidney involvement** (PMID: 30680856), and *INF2* can present with **non-FSGS histology** (minimal-change glomerulopathy, IgA nephropathy) within the same family (PMID: 29038887).

Quality-of-life impact. The combination is doubly disabling: progressive distal weakness impairs gait and manual dexterity (requiring orthoses), while progression to ESRD imposes dialysis dependence or transplantation. No formal EQ-5D/SF-36 data specific to CMTDIE are published.

Frequency note. Within *INF2*-mutation carriers, both variable penetrance and intrafamilial variability are documented—the same variant can produce isolated FSGS in one relative and the full dual phenotype in another (PMID: 25943269).

Section 4: Genetic / Molecular Information

Causal gene. *INF2* (Inverted Formin 2), 14q32.33, HGNC:23791, OMIM *610982. *INF2* is a member of the diaphanous-related formin family that nucleates and elongates actin filaments and also regulates microtubule dynamics.

Pathogenic variants. - **Domain clustering:** Nearly all CMTDIE variants localize to the **diaphanous-inhibitory domain (DID)** encoded by exons 2–4. Boyer et al. found all nine mutations in exons 2–3 (PMID: 22187985). - **Variant type:** Predominantly **missense** (e.g.,

p.L77P, p.L128P, p.G114D, p.L132P, p.G73D, p.V108D), with some **in-frame deletions** (p.Leu69_Ser72del) and **cryptic splice** variants (c.271C>G producing p.Arg91_Gln130del) (PMID: 22961558, PMID: 24174593, PMID: 24750328, PMID: 30680856). - **Classification:** Pathogenic/likely pathogenic per ACMG (segregation with disease, absence in controls, functional data). Variants are typically **germline**; *de novo* events documented. - **Allele frequency:** Essentially absent from population databases (gnomAD)—consistent with a rare, penetrant, dominant disorder. - **Functional consequence:** **Gain-of-function / dominant-negative** on actin regulation (see below), NOT loss-of-function.

Positional genotype-phenotype correlation. Ueda et al. showed that **variants between residues 184 and 245 produce isolated (monogenic) FSGS**, while **variants between residues 57 and 184 cause the dual CMT/FSGS phenotype** (PMID: 37491439). The paper states: *"Variants between residues 184 and 245 of INF2, an actin assembly factor, produce the monogenic FSGS phenotype. Meanwhile, variants between residues 57 and 184 cause a dual-faceted disease involving peripheral neurons and podocytes."* Mechanistically, CMT/FSGS variants (G73D, V108D) caused more severe cytoskeletal disruption and mitochondrial fragmentation than FSGS-only variants (T161N, N202S), providing a molecular basis for why the more N-terminal variants add the neuropathy.

Gain-of-function evidence. Subramanian et al. demonstrated that the R218Q point mutation, but not the knockout allele, confers susceptibility to glomerular disease in mice — *"R218Q INF2 mice are susceptible to glomerular disease, in contrast to INF2 knockout mice"* — and cellular assays showed the mutation alters the actin cytoskeleton via a gain-of-function effect (PMID: 39536114, PMID: 38915495). Labat-de-Hoz et al. summarize: *"These mutations disrupt INF2 regulation, leading to excessive actin polymerization"* (PMID: 39586895).

Modifier genes / epigenetics / chromosomal abnormalities. The variant *position* is the dominant modifier. No specific modifier genes, disease-specific epigenetic marks, or chromosomal abnormalities are established for CMTDIE.

Gene/GO annotations. *INF2* — GO:0007015 (actin filament organization), GO:0051017 (actin filament bundle assembly), GO:0000266 (mitochondrial fission), GO:0032956 (regulation of actin cytoskeleton organization). Cellular component: GO:0005783 (endoplasmic reticulum), GO:0005884 (actin filament).

Section 5: Environmental Information

CMTDIE is a **monogenic disease with no established environmental cause**. There are no implicated toxins, radiation, pollution, occupational exposures, lifestyle factors, or infectious agents. The only environmental dimension is experimental: in mouse models, superimposed glomerular injury (PAN, protamine sulfate) is required to unmask the renal phenotype in R218Q knock-in animals ([PMID: 38915495](#), [PMID: 27350175](#)), suggesting—by inference—that podocyte stressors may modulate onset in humans. Standard nephroprotective avoidance of nephrotoxins is prudent but not disease-specific.

Section 6: Mechanism / Pathophysiology

Ordered causal chain

1. Heterozygous missense/in-frame mutation in INF2 DID (exons 2–4)
 - | leads to
2. Loss of DID-mediated autoinhibition of INF2 (normally held inactive by a CAP1 / lysine-acetylated-actin complex bound to the DID)
 - | results in
3. Constitutive / dysregulated INF2 activity → EXCESSIVE ACTIN POLYMERIZATION (gain-of-function; NOT haploinsufficiency)
 - |
 - BRANCH A: SCHWANN CELL (peripheral nerve)

4a. Perturbation of the INF2–MAL–CDC42 myelination pathway
leads to

5a. Global disruption of the Schwann-cell actin cytoskeleton;
abnormal β -actin accumulation in Schwann cell cytoplasm
results in

6a. Chronic demyelination/remyelination + progressive axonal loss
→ intermediate nerve conduction velocities
→ CMT phenotype (distal weakness, atrophy, pes cavus)
 - BRANCH B: PODOCYTE (kidney glomerulus)

4b. Disrupted INF2 sequestration of Dynll1 → Dynll1 captured by PI3I
leads to

5b. Dynein-mediated transport of nephrin to the proteasome →
proteasome-mediated nephrin degradation → slit-diaphragm loss
(in parallel) abnormal mitochondrial fission/adhesion defects
results in

6b. Foot-process effacement, proteinuria → FSGS → ESRD (3rd–4th decade)
7. CONVERGENT TERMINAL STEP (both cell types): excess F-actin drives MRTF/SRF transcriptional reprogramming + abnormal mitochondrial dynamics → mitotic abnormalities → p53-mediated cell death → cell loss

Steps 1–3 and the podocyte branch (4b–6b) are experimentally demonstrated; the Schwann-cell branch (4a–6a) is supported by human nerve pathology and INF2–MAL interaction data but the in-vivo chain in nerve is partly inferred. Step 7 is drawn from cell-biology/review synthesis.

Detail by category

Molecular pathways. The central lesion is dysregulated **actin polymerization** by an ER-anchored formin. In Schwann cells, INF2 acts through the **INF2–MAL–CDC42** pathway; Boyer et al. showed *"INF2 colocalizes and interacts with MAL in Schwann cells. The INF2 mutants perturbed the INF2-MAL-CDC42 pathway"* (PMID: 22187985). Downstream, excess G/F-actin signaling engages the **MRTF/SRF** transcriptional axis (PMID: 39586895). INF2 is normally activated physiologically through *"calmodulin binding, KAc-actin deacetylation, G-actin binding, or association with the Cdc42 GTPase"* — regulatory inputs bypassed by DID mutations.

Cellular processes. Dysregulated actin dynamics, defective **intracellular/vesicular trafficking**, abnormal **mitochondrial fission and fusion** (INF2 nucleates actin at ER-mitochondria contact sites to drive DRP1-mediated fission), impaired cell adhesion, and ultimately **p53-mediated cell death** (PMID: 39586895, PMID: 39184068, PMID: 39774009).

Protein dysfunction. DID mutations abolish the DID–DAD (diaphanous autoregulatory domain) autoinhibitory clamp, releasing the FH2 domain to over-nucleate actin. This is a **gain-of-function**/dominant mechanism, definitively shown because point mutants but not knockouts cause disease (PMID: 39536114).

Metabolic / mitochondrial changes. Mutant INF2 causes abnormal mitochondrial dynamics and fragmentation; CMT/FSGS variants produce more severe mitochondrial fragmentation than FSGS-only variants (PMID: 37491439). Mitochondrial-associated ER membrane (MAM) actin dynamics are implicated in podocyte injury (PMID: 41864363).

Tissue damage mechanisms. Schwann cell: Mathis et al. examined six CMTDIE nerve biopsies and reported that *"these lesions reflect a global disorder of the actin cytoskeleton in Schwann cells and that CMTDIE is the first peripheral nerve disorder associated with a Schwann cell actinopathy,"* including *"abnormal accumulation of β -actin in the cytoplasm of Schwann cells"* (PMID: 24487800). Podocyte: nephrin proteostasis failure and foot-process effacement.

Biochemical abnormality (podocyte, druggable node). Sun et al. and Williquett et al. defined the mechanism: *"The R218Q mutation in INF2 disrupted sequestration of Dynll1 by INF2, allowing Dynll1 to be captured by PI31 and promoting dynein-mediated transport of nephrin to the proteasome"* (PMID: 33443052, PMID: 39621430). Proteasome inhibition (bortezomib) or knockdown of PI31/Dynll1 restored nephrin proteostasis and protected R218Q mice against PAN-induced FSGS.

Terminal transcriptional / cell-death cascade. Labat-de-Hoz et al. describe how excess actin causes *"altered intracellular trafficking, abnormal mitochondrial dynamics, and profound transcriptional reprogramming via the MRTF/SRF complex, resulting in mitotic abnormalities and p53-mediated cell death"* (PMID: 39586895).

Cell types (CL terms): Schwann cell (CL:0002573), podocyte (CL:0000653). **Anatomy (UBERON):** peripheral nerve (UBERON:0000044), renal glomerulus (UBERON:0000074). **GO biological processes:** GO:0007015 (actin filament organization), GO:0000266 (mitochondrial fission), GO:0006511 (ubiquitin-dependent protein catabolic process).

Section 7: Anatomical Structures Affected

Organ level. - Primary: Peripheral nervous system (peripheral nerves; UBERON:0000044) and kidney (renal glomerulus; UBERON:0000074). - **Body systems:** Nervous system (peripheral) and urinary/renal system. - **Secondary:** Skeletal deformities secondary to neuropathy (pes cavus, kyphoscoliosis); ESRD complications (cardiovascular, anemia, mineral-bone disease).

Tissue and cell level. - Nervous tissue: myelinating **Schwann cells** (CL:0002573) — the principal cellular target in nerve; secondary axonal loss. - **Renal tissue:** glomerular visceral epithelial cells (**podocytes**, CL:0000653) — the principal renal target; the slit diaphragm (nephrin/podocin) is the molecular casualty.

Subcellular level (GO cellular component). Endoplasmic reticulum (GO:0005783, where ER-anchored INF2 resides), actin cytoskeleton (GO:0015629), mitochondrion (GO:0005739, abnormal fission/fusion), proteasome complex (GO:0000502, nephrin degradation), ER-mitochondria contact site / MAM.

Localization / lateralization. Neuropathy is **bilateral, symmetric, length-dependent** (distal legs first). Renal involvement is bilateral (systemic glomerular disease).

Section 8: Temporal Development

Onset. Neuropathy typically begins in **childhood to adolescence** (mean CMT onset ~11.5 years, range 3–17) with insidious, slowly progressive distal weakness (PMID: 30680856). Renal onset is variable, ranging from childhood to adulthood (PMID: 23014460).

Progression. Both components are **chronic and progressive**. Neuropathy progresses slowly over decades. Renal disease progresses from proteinuria to nephrotic syndrome to FSGS: Barua et al. report that *"INF2-related disease showed variable penetrance, with onset of disease ranging widely from childhood to adulthood, and commonly leading to end-stage renal disease in the third and fourth decade of life"* (PMID: 23014460). Renal dysfunction is more severe and earlier-onset when neuropathy coexists (PMID: 24174593).

Patterns. No spontaneous remission. Disease is lifelong. The window for renal intervention is before advanced glomerulosclerosis; the theoretical critical period for any future INF2-directed therapy would be prior to irreversible podocyte loss.

Section 9: Inheritance and Population

Epidemiology. CMTDIE is **rare** (no precise prevalence; part of the broader CMT spectrum affecting ~1 in 2,500). Among **autosomal-dominant familial FSGS**, *INF2* mutations explain ~9%: Barua et al. found *"Mutations in INF2 were found in a total of 20 of the 215 families... thereby explaining disease in 9%"* versus only 2/281 sporadic cases (PMID: 23014460). By comparison, *ACTN4* accounted for ~3% and *TRPC6* ~2%.

Inheritance genetics. - **Pattern:** Autosomal dominant; *de novo* mutations documented (PMID: 24174593). - **Penetrance:** Variable/incomplete — "variable penetrance, with onset ranging widely from childhood to adulthood" (PMID: 23014460). - **Expressivity:** Variable, including intrafamilial variability where the same variant causes isolated FSGS in one relative and dual CMT/FSGS in another (PMID: 25943269). - **Anticipation, mosaicism, founder effects, consanguinity:** Not established (dominant, non-repeat-expansion disease; consanguinity not relevant).

Population demographics. Reported across diverse populations—European, Korean (PMID: 24750328), Chinese (PMID: 25943269, PMID: 31515790)—with no ethnic predilection. No strong sex bias reported for the Mendelian disease.

Section 10: Diagnostics

Clinical tests. - **Electrophysiology (key):** Nerve conduction studies show **intermediate motor NCV** with both demyelinating and axonal features—the diagnostic signature ([PMID: 24750328](#)). - **Urinalysis / renal labs:** Proteinuria screening is essential in every CMT patient — “we strongly suggest to screen for proteinuria in CMT patients, in order to identify patients with this renal-neurologic phenotype in an early stage” ([PMID: 25439738](#)). Serum albumin, creatinine, eGFR track renal function. - **Nerve biopsy (sural):** Chronic demyelination/remyelination, progressive axonal loss, whorl-like Schwann-cell proliferations, abnormal β -actin accumulation — a Schwann-cell actinopathy ([PMID: 24487800](#)). - **Renal biopsy:** FSGS (or occasionally minimal-change/IgA histology) ([PMID: 29038887](#)).

Genetic testing (definitive). Targeted **single-gene *INF2* sequencing (exons 2–4)** or **CMT/FSGS gene panels; whole-exome sequencing** has identified novel variants (e.g., p.L132P) ([PMID: 24750328](#)). Screening should not be restricted to patients with combined neuro-renal disease, since *INF2* variants can cause isolated CMT ([PMID: 30680856](#)) or isolated FSGS. *INF2* testing is strongly recommended in any patient with CMT plus early nephropathy ([PMID: 24174593](#)). Diagnosis via genetics can sometimes obviate renal biopsy ([PMID: 27733133](#)).

Clinical criteria / differential diagnosis. Differentiate from other intermediate CMTs (CMT1X/*GJB1*, DI-CMT from *DNM2*, *YARS*), CMT1A (*PMP22* duplication), and isolated genetic FSGS (*NPHS2*, *TRPC6*, *ACTN4*, *WT1*). The combination of intermediate NCV **plus** proteinuria/FSGS strongly points to *INF2*.

Screening. Cascade genetic testing of at-risk relatives; urine protein screening in known carriers. No newborn screening exists.

Section 11: Outcome / Prognosis

Renal outcome. Progressive to ESRD, “commonly leading to end-stage renal disease in the third and fourth decade of life” ([PMID: 23014460](#)). Renal disease is generally steroid-resistant.

Key transplant prognostic distinction. Genetic (*INF2*) FSGS **does not recur** after kidney transplantation, in sharp contrast to idiopathic FSGS: “Whilst patients with FSGS without a confirmed genetic cause have a high recurrence rate in the transplanted organ, patients with a mutation generally exhibit no recurrence and have a good prognosis” ([PMID: 27733133](#)). Direct

clinical confirmation: in an *INF2* family with 14 affected members, "Four members received a kidney transplant without disease recurrence" (PMID: 29038887). This is a **major counseling anchor**—transplantation offers durable renal replacement.

Neurologic outcome. Slowly progressive disability from distal weakness, atrophy, and foot deformity; not typically life-limiting on its own. Life expectancy is governed largely by renal outcome and transplant success.

Prognostic factors. Variant position (dual vs isolated phenotype), age at renal onset, degree of proteinuria/glomerulosclerosis at diagnosis.

Section 12: Treatment

No disease-specific/curative therapy currently exists. Management is supportive and organ-directed.

Neurologic / supportive-rehabilitative. - Physical therapy, occupational therapy, ankle-foot orthoses, orthopedic management of pes cavus/kyphoscoliosis (NCIT: Physical Therapy, Orthotic Device). Symptomatic pain management as needed.

Renal. - RAAS blockade (ACE inhibitors/ARBs) for proteinuria (antiproteinuric, nephroprotective; NCIT: ACE Inhibitor, Angiotensin Receptor Antagonist). FSGS here is generally steroid-resistant, so immunosuppression is of limited value. - **Renal replacement:** dialysis and **kidney transplantation** (NCIT: Kidney Transplantation)—the latter with excellent, non-recurring outcomes (PMID: 27733133).

Experimental / emerging (mechanism-directed). - **Proteasome inhibition:** Bortezomib restored nephrin proteostasis and protected R218Q mice — "*Suppression of proteasome-mediated proteolysis with proteasome inhibitors is a new therapeutic strategy for inverted formin 2-mediated FSGS*" (PMID: 39621430). Targeting the PI31–Dyln11 interaction is a proposed node. - **Allele-selective silencing (ASO/siRNA):** Because knockout is non-pathogenic while the point mutant is, selectively silencing the mutant allele is a rational (untested-in-human) strategy. - No pharmacogenomic, gene-therapy, cell-therapy, or immunotherapy protocols are established for CMTDIE.

Treatment strategy. Genotype-guided: confirm *INF2* variant, monitor proteinuria and nerve function, initiate RAAS blockade early, plan for transplantation, and counsel on non-recurrence.

Section 13: Prevention

Primary prevention. Not possible for a germline Mendelian disorder. **Genetic counseling** is central: autosomal dominant inheritance means 50% transmission risk; prenatal diagnosis and **preimplantation genetic diagnosis (PGD)** are options for known family variants.

Secondary prevention. **Cascade genetic testing** of at-risk relatives and **proteinuria screening** in carriers enables early detection and early RAAS blockade to slow renal progression ([PMID: 25439738](#)).

Tertiary prevention. Nephroprotection (blood-pressure control, avoidance of nephrotoxins), management of ESRD complications, orthopedic/rehabilitative care to preserve mobility, and timely transplantation.

Immunization / public health / environmental interventions. Not applicable (non-infectious, non-environmental).

Section 14: Other Species / Natural Disease

- **Taxonomy:** *INF2* orthologs exist in mammals; disease modeling is chiefly in **mouse (*Mus musculus*, NCBI Taxon 10090)**.
 - **Orthologous gene:** mouse *Inf2* (NCBI Gene ID 79600).
 - **Natural disease:** No well-characterized naturally occurring CMTDIE in companion animals or wildlife is documented (OMIA). The disease is essentially known from humans and engineered models.
 - **Comparative biology:** *INF2*'s actin-regulatory and mitochondrial-fission functions are evolutionarily conserved, supporting cross-species mechanistic relevance.
 - **Transmission:** Not applicable (non-communicable, germline genetic).
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Section 15: Model Organisms

Mouse models (principal).

Model	Type	Key finding	PMID
<i>Inf2</i> R218Q knock-in	Point-mutant knock-in	Susceptible to PAN-induced proteinuria/FSGS; demonstrates gain-of-function	39536114 , 38915495
<i>Inf2</i> knockout	Null allele	Minimal renal phenotype — does NOT recapitulate disease	39536114
<i>Inf2</i> R218Q knock-in (protamine)	Injury model	Impaired podocyte/slit-diaphragm recovery; nephrin/podocin mislocalization	27350175
Patient iPSC kidney organoid (S186P)	In vitro human	Recapitulates defective adhesion and mitochondrial phenotypes	38915495

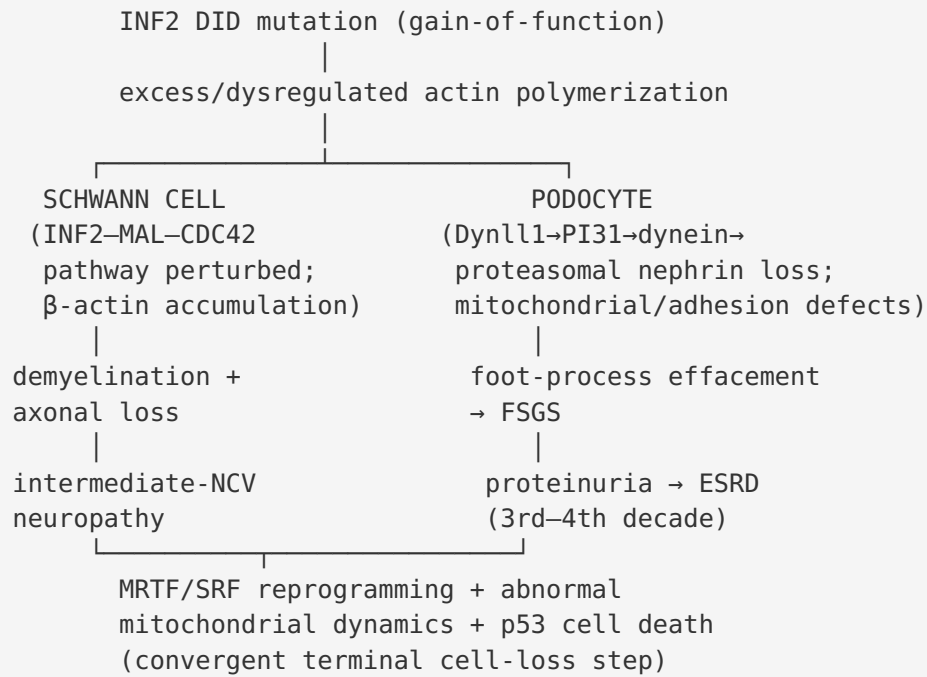
Phenotype recapitulation. The R218Q knock-in reproduces the **renal** phenotype (only after a second-hit injury) and demonstrates the gain-of-function mechanism and the therapeutic tractability of proteasome inhibition. Human iPSC-derived podocyte organoids recapitulate the adhesion/mitochondrial defects.

Model limitations. (1) The **neuropathy** component is poorly modeled—published mouse work focuses on kidney, not Schwann-cell disease. (2) Baseline mice are near-normal; a stressor is required to unmask renal disease, so the models capture *susceptibility* rather than spontaneous progressive FSGS. (3) The positional genotype-phenotype gradient (why some variants add neuropathy) is not fully reconstructed in vivo.

Resources: MGI (mouse *Inf2*), patient-derived iPSC/organoid lines.

Mechanistic Model / Interpretation

CMTDIE is best understood as a **single molecular lesion producing a two-organ actinopathy**. A DID mutation releases INF2 from autoinhibition, and the resulting excess/dysregulated actin polymerization is simultaneously toxic to the two most architecturally demanding cell types in the body—myelinating Schwann cells and podocytes—both of which depend on exquisitely controlled actin cytoskeletons for their elaborate membrane processes (myelin wraps; foot processes/slit diaphragm).



The **positional gradient** (residues 57–184 → dual; 184–245 → renal-only) implies that the N-terminal DID region governs an interaction (plausibly the Schwann-cell INF2–MAL–CDC42 axis) whose disruption is required to add neuropathy, whereas podocyte injury is triggered across a broader mutational span. The **gain-of-function** nature reframes therapy: rather than replacing lost function, the goal is to *reduce* aberrant activity—hence the appeal of allele-selective silencing (mimicking the benign knockout) and downstream proteasome inhibition (rescuing nephrin).

Evidence Base

PMID	Title (abbrev.)	Contribution
22187985	<i>INF2 mutations in CMT with glomerulopathy</i>	Establishes INF2 DID as cause in 75% of CMT+FSGS; INF2–MAL–CDC42 in Schwann cells
37491439	<i>Cytoskeletal/structural effects of INF2 variants</i>	Positional genotype-phenotype gradient (57–184 dual; 184–245 renal-only)
39536114	<i>INF2 causes kidney disease through gain-of-function</i>	Point-mutant but not knockout causes disease → gain-of-function
38915495	<i>Missense mutant gain-of-function INF2-FSGS</i>	R218Q knock-in + organoid recapitulate adhesion/ mitochondrial defects
24487800	<i>Neuropathology: Schwann cell actinopathy</i>	Defines nerve pathology as Schwann-cell actinopathy; β -actin accumulation
23014460	<i>INF2 in familial vs sporadic FSGS</i>	INF2 = 9% of AD familial FSGS; variable penetrance; ESRD in 3rd–4th decade
33443052	<i>Dysregulated dynein trafficking of nephrin</i>	Podocyte mechanism: nephrin mistrafficking
39621430	<i>Dynll1-PI31 / proteasome target</i>	Dynll1→PI31→proteasomal nephrin loss; proteasome inhibition therapeutic
39586895	<i>Regulation of INF2 in inherited disorders</i>	Excess actin → MRTF/SRF + p53 cell death; physiological activators
27733133	<i>Diagnosing FSGS without biopsy</i>	Genetic FSGS does not recur post-transplant
29038887	<i>INF2 with non-FSGS histology</i>	4 transplants without recurrence; histologic heterogeneity
30680856	<i>Cryptic splice INF2, minimal renal</i>	INF2 can cause isolated CMT; expands testing indications
24174593	<i>De novo INF2 mutations</i>	De novo events; broader phenotype (ID, hearing loss)
25439738	<i>CMT: are you testing for proteinuria?</i>	Clinical mandate to screen CMT patients for proteinuria

Evidence source types: Human clinical/genetic (case series, pedigrees, cohort genotyping); model organism (R218Q knock-in and knockout mice); in vitro (patient iPSC organoids, cultured podocytes); computational/structural (variant modeling).

Limitations and Knowledge Gaps

1. **Neuropathy mechanism underexplored in vivo.** Mouse work centers on kidney; no robust mouse model reproduces the Schwann-cell neuropathy, leaving the INF2–MAL–CDC42 chain partly inferential.
 2. **Positional gradient not mechanistically closed.** Why residues 57–184 add neuropathy while 184–245 spare nerve is correlative; the specific N-terminal interaction responsible is not proven.
 3. **Epidemiology imprecise.** No population-level prevalence/incidence for CMTDIE specifically; frequency is anchored to familial-FSGS cohorts.
 4. **Penetrance/expressivity unexplained.** Variable and intrafamilial variability lack identified modifiers (genetic or environmental).
 5. **No human therapeutic data.** Proteasome inhibition and allele-selective silencing are preclinical; efficacy/safety in patients is unknown.
 6. **Second-hit requirement in models** complicates translation—human triggers of renal onset are not defined.
 7. **QoL data absent.** No formal EQ-5D/SF-36/PROMIS metrics for the dual disability.
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Proposed Follow-up Experiments / Actions

1. **Develop a neuropathy-competent model** — Schwann-cell-specific R218Q/G73D knock-in mice or patient iPSC-derived Schwann cells/organoids to test the INF2–MAL–CDC42 hypothesis and screen neuroprotective compounds.
2. **Test allele-selective silencing** (ASO/siRNA against the mutant *INF2* allele) in R218Q knock-in mice for both renal and (once modeled) neural readouts, leveraging the benign-knockout rationale.

3. **Advance proteasome-axis therapeutics** — dose-ranging bortezomib and PI31–Dynll1 interaction inhibitors in knock-in models with proteinuria and nephrin proteostasis endpoints.
4. **Structure-function mapping** of DID residues 57–245 to explain the phenotype gradient (co-IP/proximity assays for MAL/CDC42/Dynll1 binding across variants).
5. **Build a CMTDIE patient registry** capturing genotype, NCV, renal trajectory, transplant outcomes, and QoL to quantify penetrance, ESRD timing, and non-recurrence rates prospectively.
6. **Modifier discovery** — WGS + expression profiling in variable-expressivity families to find genetic/environmental modifiers of neural vs renal severity.
7. **Clinical guideline** — codify universal proteinuria screening in CMT and *INF2* testing in isolated CMT or FSGS, plus counseling on transplant non-recurrence.

Report compiled from 8 confirmed findings and 27 reviewed papers across 5 investigation iterations. Ontology suggestions: MONDO:0013758; genes/GO: INF2 (GO:0007015, GO:0000266, GO:0005783); cells CL:0002573 (Schwann cell), CL:0000653 (podocyte); anatomy UBERON:0000044 (peripheral nerve), UBERON:0000074 (renal glomerulus); phenotypes HP:0000097 (FSGS), HP:0000093 (proteinuria), HP:0001761 (pes cavus), HP:0003774 (ESRD).
